

Data Science

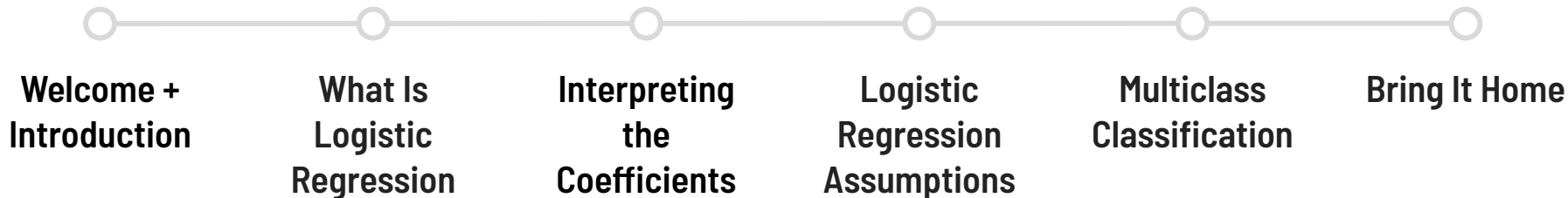
Logistic Regression

Machine Learning Fundamentals



GENERAL ASSEMBLY

Lesson roadmap



Learning objectives



1

Describe the principles of logistic regression.

2

Describe how logistic regression is similar to and different from linear regression.

3

Interpret the coefficients of a logistic regression model.

4

State the assumptions of a logistic regression model.

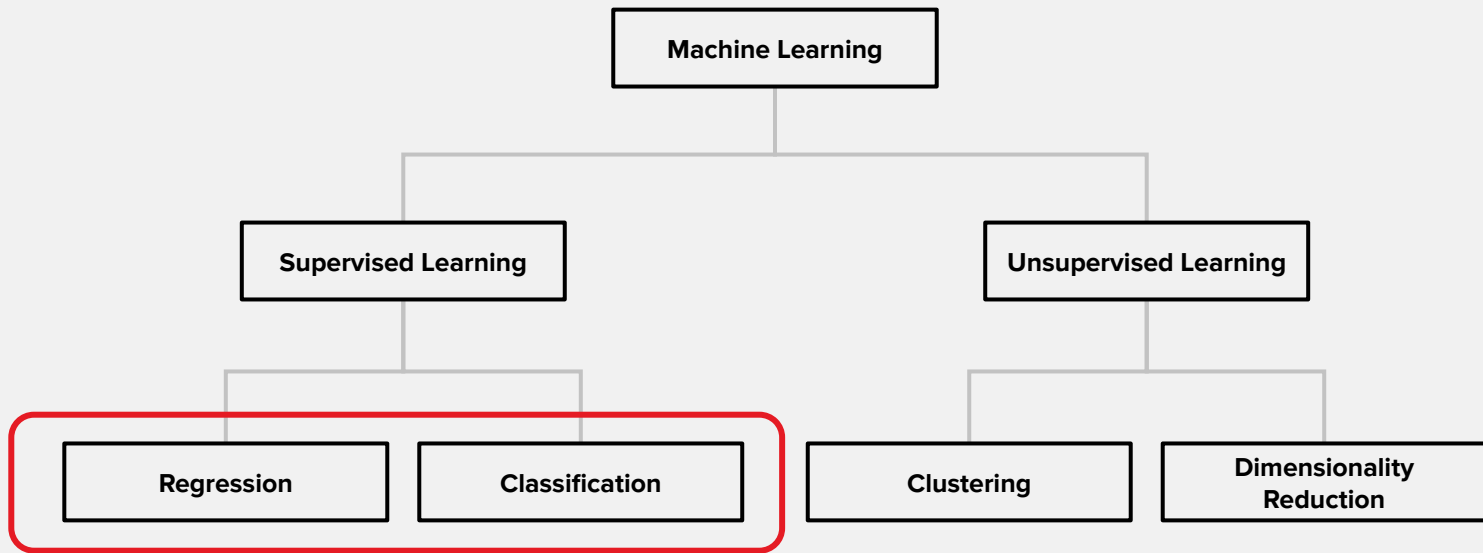
5

Fit, evaluate, and generate predictions from a logistic regression model in `scikit-learn`.

What is Logistic Regression?

Supervised, white-box, regression

Where are we now?



We are here!

Reminder: Two Kinds of Supervised Learning

Regression - this is when our y-variable is numeric.

- *“Given the past values of the stock price of Apple, what will tomorrow’s closing price be?”*
- *“Given the annual precipitation, average temperature, and soil pH, what will this year’s harvest yield be?”*
- *“Given the square footage, number of bedrooms, number of bathrooms, and quality of school district, what will the price of this home be?”*

Reminder: Two Kinds of Supervised Learning

Classification - this is when our y-variable is a category. If it's a 0/1 yes/no kind of variable, we often call it **binary classification**. Otherwise, **multiclass classification**.

- *“Given this person’s demographic information, how many tabs they have open, and where they live, will they make a purchase on my site?”*
- *“Given radar readouts, past weather, and almanac data, will it rain tomorrow?”*
- *“Given how many hours you study, how many hours you sleep, and your course load, will you pass the final exam?”*

Reminder: Linear Regression

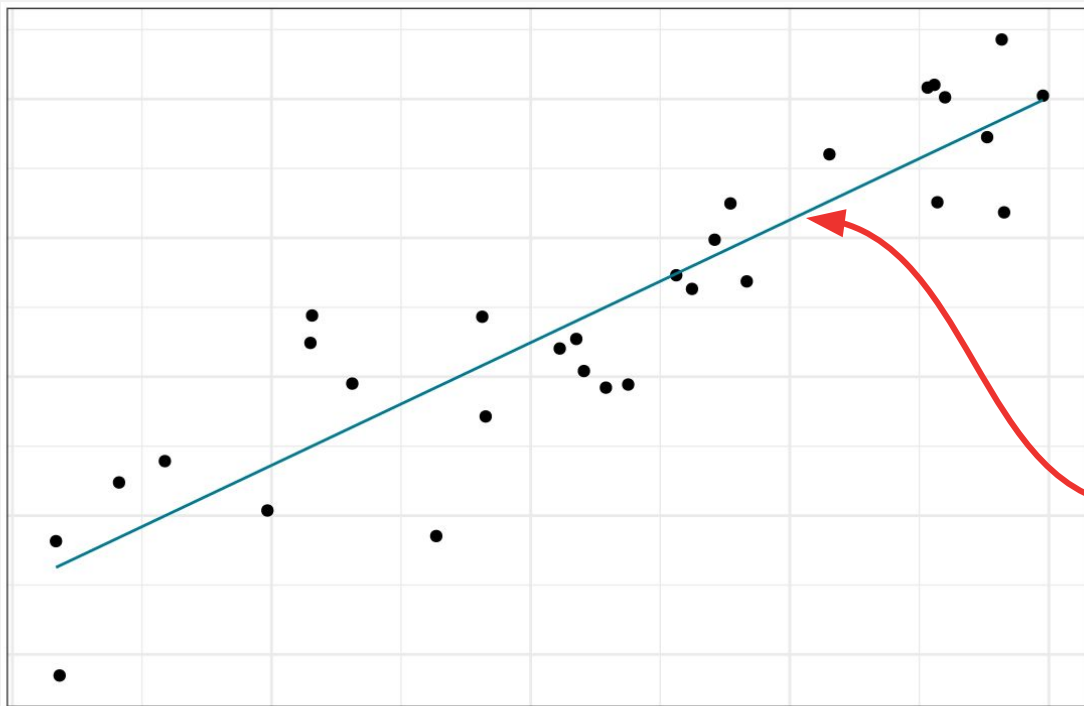
In ordinary least squares linear regression (often just referred to as **OLS**), we try to predict some response variable (y) from at least one independent variable (x). We believe there is a **linear** relationship between the two:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

Graphically, we draw a line to represent the relationship

House sale price
(y)

Continuous
variable

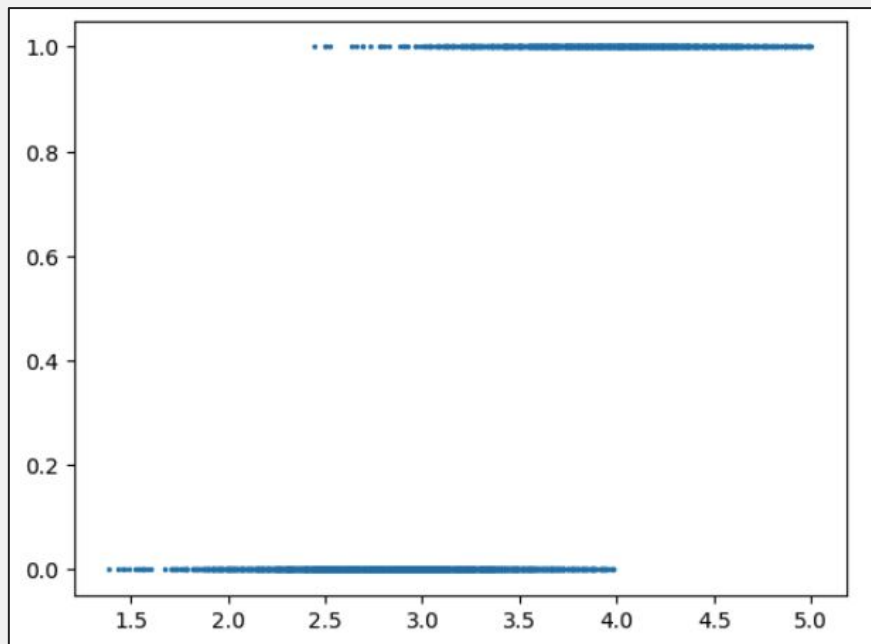


This is our
model

But what if our target is categorical...?

Admitted to college
(y)

Categorical
variable

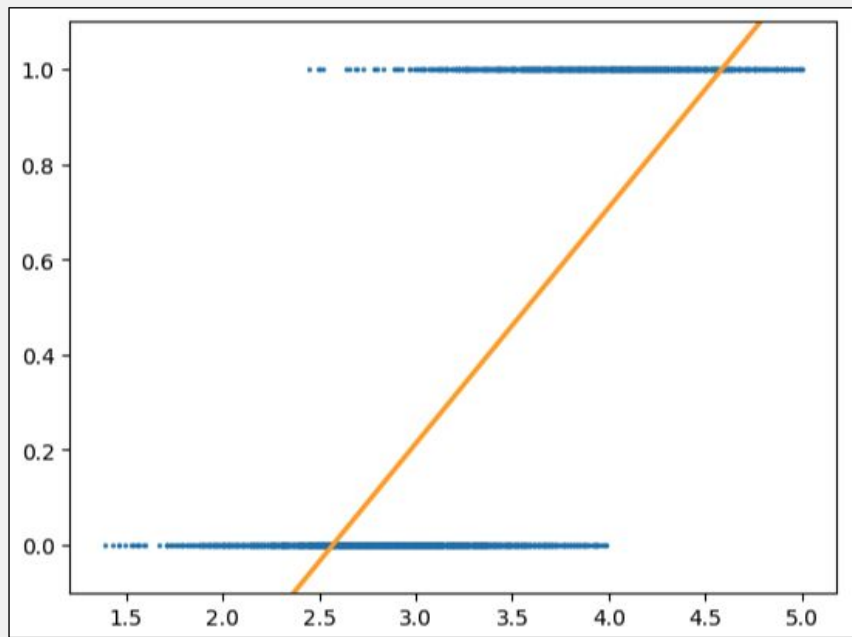


GPA
(x)

But what if our target is categorical...?

Admitted to college
(y)

Categorical
variable



GPA
(x)

We need something else

A linear regression model can predict any value from negative infinity to infinity ($-\infty$ to ∞), but we only want a value of 0 or 1!

How can we predict whether an observation is 0 (not admitted to college) or 1 (admitted to college)?

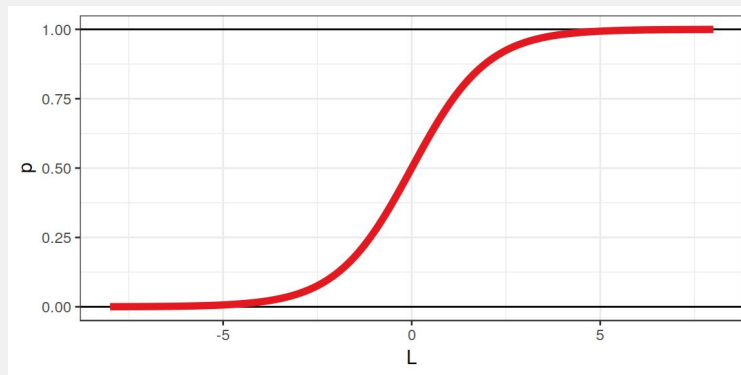


A different model

We need a model that gives outputs between 0 and 1. One of these is the **logistic function**.

Since our classes are 0 and 1, and the logistic function gives values between 0 and 1, we can work out the **probability** of an observation (GPA score) belonging to class 1 (admitted to college).

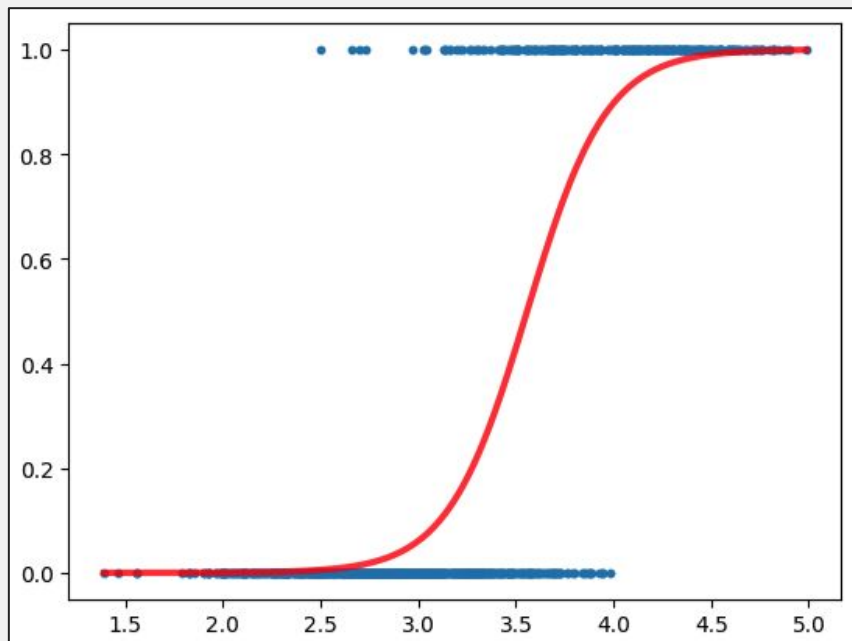
Then we can assign an observation to a class based on its probability.



The logistic function is S-shaped, allowing us to predict a class based on probabilities

Admitted to college
(y)

Categorical
variable



GPA
(x)

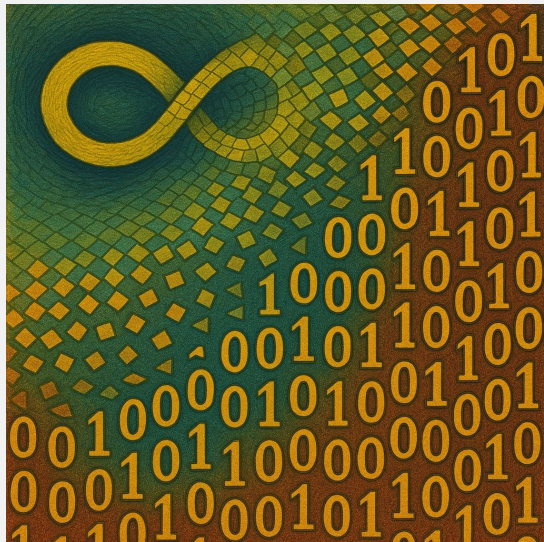
So we won't use regression any more...?

Actually we still apply a form of regression.

We can use a **link function** to effectively "bend" our line of best fit into a curve of best fit that matches the range or set of values in which we're interested (0 to 1).

For logistic regression, the link function that transforms ("bends") our line is known as the **logit link function**.

This enables us to use a regression model to work out a probability based on an input variable.



How does the logit link function work?

The logit link function transforms a probability (which is between 0 and 1) into the log-odds (which is a value on the real number line, $-\infty$ to ∞).

The log-odds are the ratio of the probability of an event happening to the probability of it not happening.

Linear Regression

$$y = \beta_0 + \beta_1 x_i + \varepsilon$$

Apply the logit link
function...

$$\text{logit}(P(y = 1)) = \beta_0 + \beta_1 X_1 + \varepsilon$$

...which predicts the
log-odds of success

$$\ln \left(\frac{P(y = 1)}{1 - P(y = 1)} \right) = \beta_0 + \beta_1 X_1 + \varepsilon$$

Logistic Regression

Let's run a Logistic Regression model!

- Open **02-logistic-regression.ipynb**



Interpreting the Coefficients

The Math Behind the Model

We interpret the coefficients in a logistic regression model the same way we do in linear regression

A one-unit change in a coefficient implies a one unit change in the log-odds of success.

$$\ln \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1 x$$

We interpret the coefficients in a logistic regression model the same way we do in linear regression

A one-unit change in a coefficient implies a one unit change in the log-odds of success.

$$\underbrace{\ln \left(\frac{p}{1-p} \right)}_{\text{log-odds}} = \underbrace{\beta_0}_{\text{Intercept}} + \underbrace{\beta_1}_{\text{Coefficient}} \underbrace{x}_{\text{Feature}}$$

But what are the log-odds?

First, let's look at odds

Scenario

You run a company that is trying to understand **how likely a returning customer is to purchase a product compared to a new customer**. Your boss wants to predict whether someone will make a purchase and you'll use past data to do it.

Past data

Makes a Purchase	New Customer	Returning Customer	Total
Yes	2	22	24
No	15	5	20
Total	17	27	44

What is the likelihood of someone making a purchase based on past data?

Past data

Makes a Purchase	New Customer	Returning Customer	Total
Yes	2	22	24
No	15	5	20
Total	17	27	44

What is the likelihood of someone making a purchase based on past data? 55%

Past data

Makes a Purchase	New Customer	Returning Customer	Total
Yes	2	22	24
No	15	5	20
Total	17	27	44

$$\text{probability of purchase} = \frac{\text{number of customers who have purchased}}{\text{all customers}} = \frac{24 \text{ (Yes)}}{44 \text{ (Total)}} = 0.545$$

What are the odds of anyone making a purchase?

The **odds** of an event occurring are:

$$\frac{\text{prob. of the event occurring}}{\text{prob. it does not occur}} = \frac{p}{1-p}$$

What are the odds of anyone making a purchase?

For example, what are the odds of rolling 1 on a fair 6-sided die?

$$\text{odds} = \frac{1/5}{5/6} = \frac{1}{5} = 5 : 1$$

What are the odds of anyone making a purchase?

For example, what are the odds of rolling 1 on a fair 6-sided die?

$$\text{odds} = \frac{1/5}{5/6} = \frac{1}{5} = 5 : 1$$


Gamblers prefer a colon with the bigger number first because it represents their payoff

What are the odds of purchase if someone is a returning customer?

Past data

Makes a Purchase	New Customer	Returning Customer	Total
Yes	2	22	24
No	15	5	20
Total	17	27	44

$$\text{odds of purchase if you are a RETURNING customer} = \frac{22(Yes)}{5(No)} = 4.4$$

What are the odds of a purchase for new customers compared with the odds of a purchase for returning customers ?

Makes a Purchase	New Customer	Returning Customer	Total
Yes	2	22	24
No	15	5	20
Total	17	27	44

$$\text{odds ratio} = \frac{\text{odds of purchase for returning customers}}{\text{odds of purchase for new customers}} = \frac{\frac{22}{5}}{\frac{2}{15}} = \frac{22 * 15}{5 * 2} = 33$$

Odds Ratio

This means that returning customers have 33 times greater odds of purchasing something than new customers.

Makes a Purchase	New Customer	Returning Customer	Total
Yes	2	22	24
No	15	5	20
Total	17	27	44

$$\text{odds ratio} = \frac{\text{odds of purchase for returning customers} - \frac{22}{5}}{\text{odds of purchase for new customers} - \frac{2}{15}} = \frac{22 * 15}{5 * 2} = 33$$

Converting from odds to probability

Odds can be tricky to interpret for non-technical audiences. You can convert from odds to probability (and vice versa)using the formula below.

$$\text{odds} = \frac{\text{probability}}{1 - \text{probability}} = \frac{p}{1 - p}$$

Probability or risk = $\frac{p}{p+q}$ 

Odds = $p : q$ 

$$\text{odds ratio} = \frac{\left(\frac{p}{1-p} \right)}{\left(\frac{p_2}{1-p_2} \right)}$$

Where does this get us?

Using some algebra, we can actually rearrange

$$\ln \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1 x$$

Logit function
(log-odds of success)

into

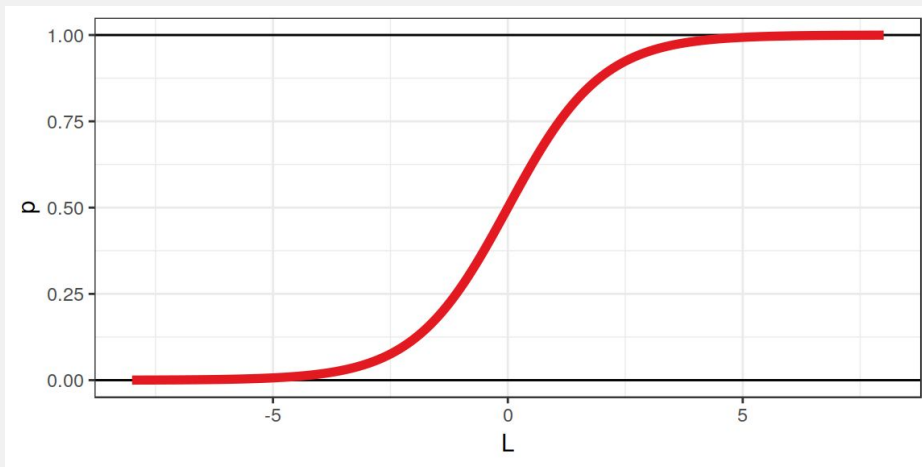
$$p = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}$$

Logistic function
(probability of success)

Great!

So, given our estimates for β_0 and β_1 , we can then compute the probability of a given event occurring based on some x !

$$p = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}} = \frac{e^L}{1 + e^L}$$



Interpreting the coefficients

Let's see it for ourselves!

- Open **02-logistic-regression.ipynb**



Logistic Regression Assumptions

Logistic Regression Assumptions

Logistic regression follows similar assumptions to OLS

- **L** - Linearity. Relationship between independent variables and log-odds y should be approximately linear.
- **I** - Independence. Your observations should not affect one another.
- **M** - No Multicollinearity/Independence of Predictors (this is for multiple logistic regression only).

Multiclass Classification

What if we want to predict more than two classes?

Logistic regression is only for binary classification problems! However, there are two "hacks" that allow us to fit multiclass classification problems using the same/similar methodology. They are:

1. **Multinomial logistic regression:** available in **scikit-learn**'s **LogisticRegression()** model
2. **One-vs-Rest (OVR) logistic regression:** available as a separate **OneVsRestClassifier** in **scikit-learn**



Bring It Home



Trivia

What is $p/1-p$?

Select the correct response

A

Probability

B

Odds

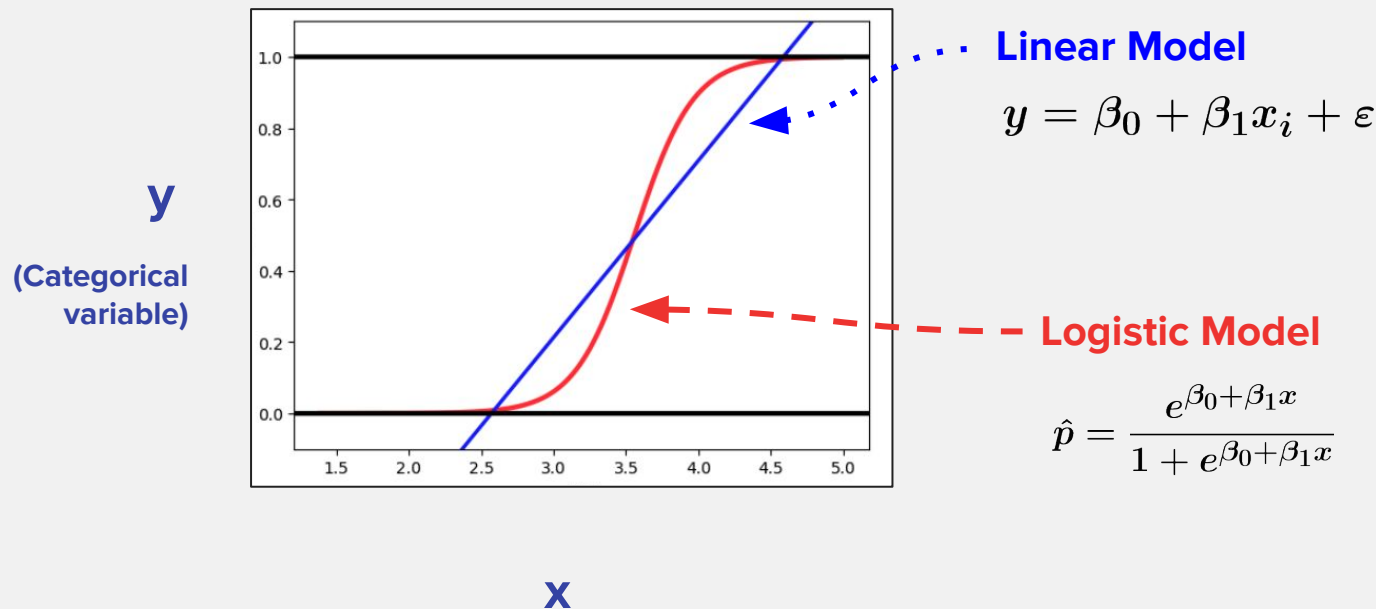
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Odds Ratio

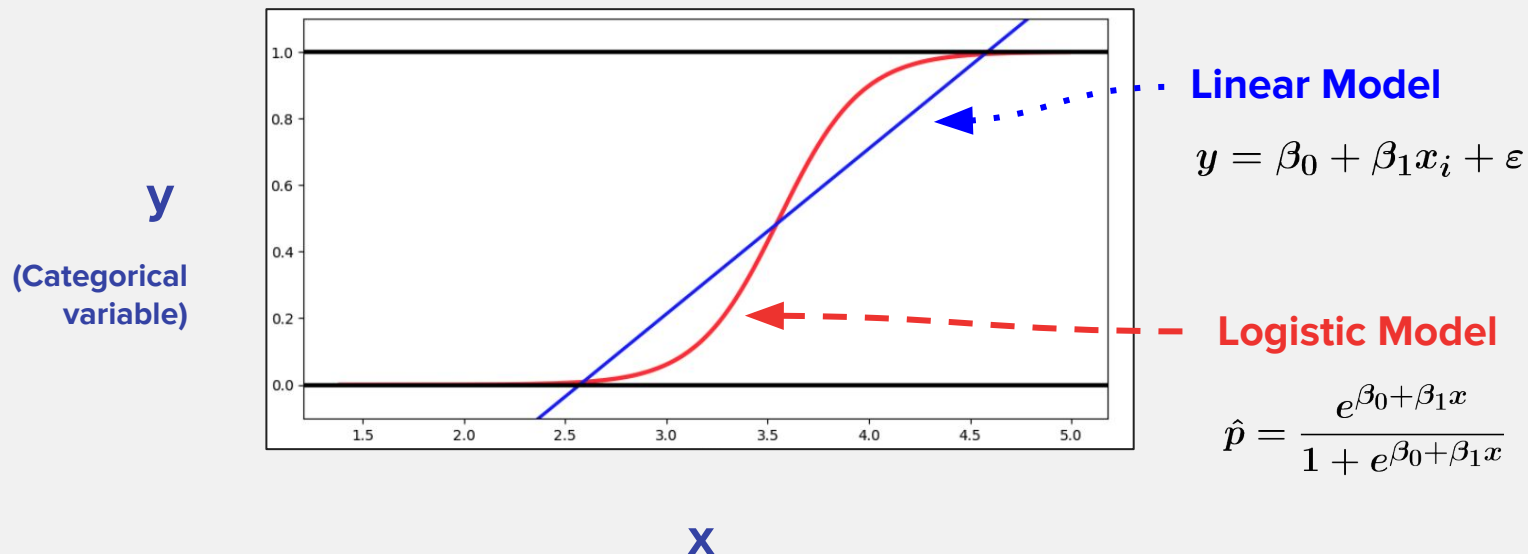
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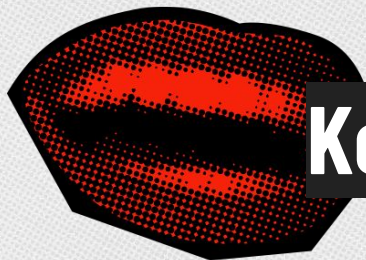
None of the above

Logistic regression enables us to predict the probability of class membership



Logistic regression enables us to predict the probability of class membership





Key takeaways

Logistic Regression

1 Strengths

Very easy to interpret

The impact of each variable on the target is quantified (the coefficients are interpretable)

2 Weaknesses

Lots of assumptions need to be met (LINEM)

3 Metrics

ROC AUC
Accuracy
F1 score

4 When use

Predicting binary outcomes

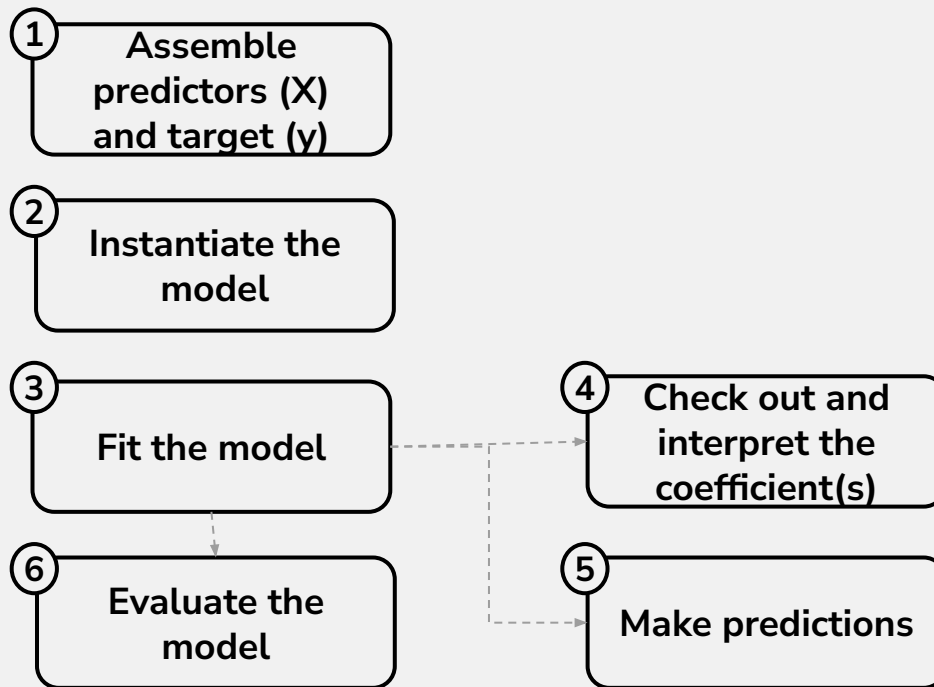
The core steps to take for Logistic Regression

```
from sklearn.linear_model import  
LogisticRegression  
X = (must be df)  
y = (must be Series)  
lm = LogisticRegression()
```

```
lm.fit(X_train, y_train)
```

```
accuracy= lm.score
```

Use classification metrics
Accuracy, ROCAUC, F1 Score



$$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x$$
$$p = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}$$

```
lm.predict()
```

